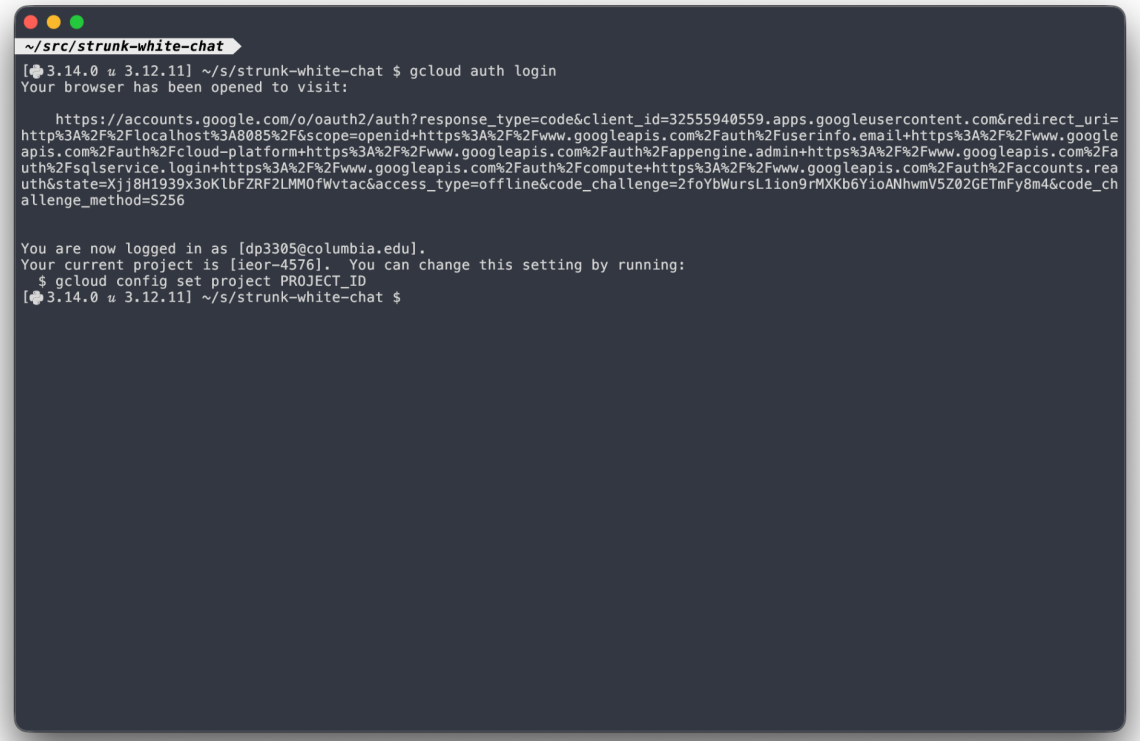
Deploying a FastAPI App to Cloud Run with Cloud Build & uv

Prerequisites

- Working `app.py` (FastAPI server)
- `pyproject.toml` with dependencies
- `uv.lock` generated with dependencies
- `gcloud` CLI [installed and configured for your project]

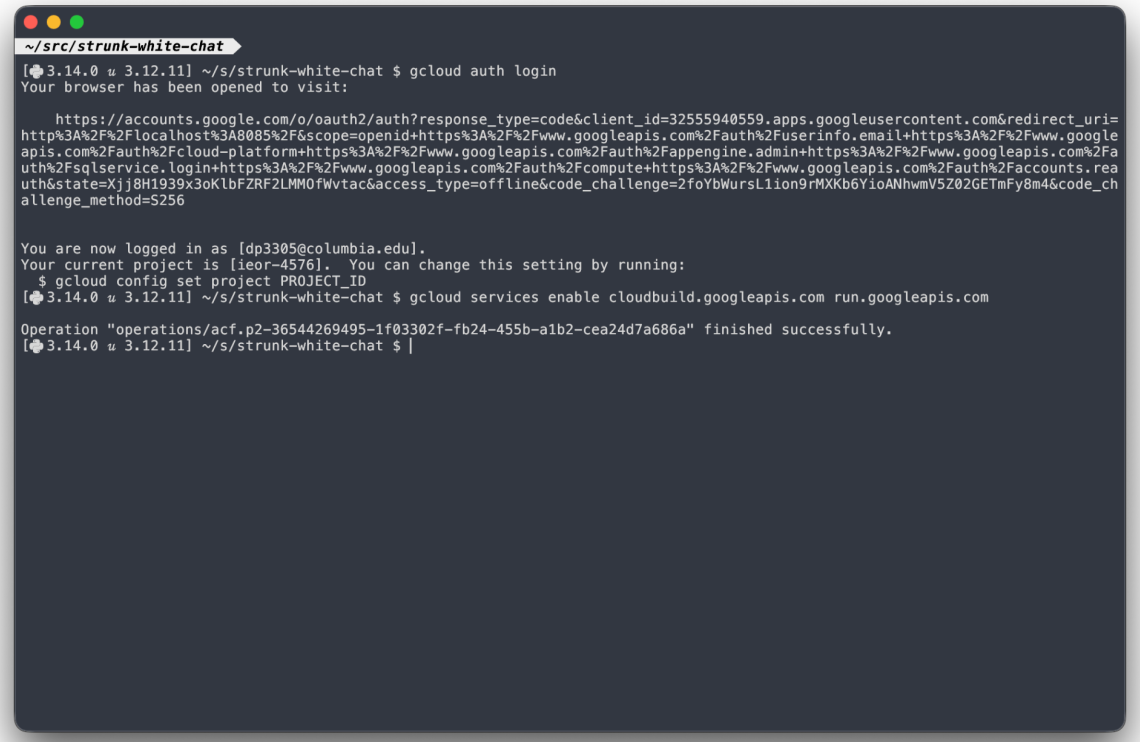
Step 1: Authenticate your `gcloud` CLI

```shell
gcloud auth login
```



Step 2: Enable the required APIs

```shell
gcloud services enable cloudbuild.googleapis.com
run.googleapis.com
```



Step 3: Add a `Dockerfile` to your project

Copy this into a file called `Dockerfile` in the root directory of your project:

```none
FROM python:3.12-slim

COPY --from=ghcr.io/astral-sh/uv:latest /uv /uvx /bin/

WORKDIR /app

COPY pyproject.toml uv.lock ./
```

```
RUN uv sync --frozen --no-dev

COPY . .

CMD ["uv", "run", "uvicorn", "app:app", "--host", "0.0.0.0", "--port", "8080"]
```

Step 4: Add a `cloudbuild.yaml` to your project

Copy this into a file called `cloudbuild.yaml` in the root directory of your project:

REPLACE `my-app` (3 places) with your application name.

```
None
steps:
  - name: 'gcr.io/cloud-builders/docker'
    args: ['build', '-t', 'gcr.io/$PROJECT_ID/my-app', '.']

  - name: 'gcr.io/cloud-builders/docker'
    args: ['push', 'gcr.io/$PROJECT_ID/my-app']

  - name: 'gcr.io/cloud-builders/gcloud'
    args:
      - 'run'
      - 'deploy'
      - 'my-app'
      - '--image=gcr.io/$PROJECT_ID/my-app'
      - '--region=us-central1'
      - '--allow-unauthenticated'
      - '--port=8000'
```

Step 5: Add a `.gcloudignore`

This is important to reduce the deployment size and speed up the build process. Add this next to the `Dockerfile` and `cloudbuild.yaml`

```
None

.venv
```

```
__pycache__
.pytest_cache
.git
```

Step 6: Deploy

This final step is what you'll be doing as your build and

```Shell
gcloud builds submit .
```